Realistic Loss Budgeting and Defect Tolerance in Lead-Free $RbGeI_3$ Thin-Film Solar Cells: A SCAPS-1D and SHAP Sensitivity Analysis

Md. Abdul Malek Fahim[a], Hussain Touhid Siddiquee[a,*]

[a] *Department of Electrical and Electronic Engineering, Leading University, Sylhet, Bangladesh*

Abstract

Single-junction SCAPS-1D optimizations of $RbGeI_3$ perovskite cells have converged on 25–31% idealized efficiencies, leaving editors and experimentalists asking what survives contact with reality and what is structurally new. We keep the low-cost all-inorganic FTO/TiO_2/$RbGeI_3$/CuI/Au stack and add three elements missing from its literature: (i) a Cu_2O-like p^+ back-surface field (BSF) that raises the ideal-coupling point from 26.81% (V_{OC} 1.13 V, J_{SC} 30.32 mA/cm^2, FF 78.26%) to 30.35% (1.18 V, 30.32 mA/cm^2, 84.89%), with doping, not thickness, controlling the gain; (ii) a loss waterfall — native defect/radiative/Auger simulation plus photon-scaling optics and post-processed parasitics — that bounds the experimentally plausible window at 20.8% without BSF and 23.3% with BSF on the identical defect stack; and (iii) a data-driven global sensitivity analysis — an XGBoost surrogate with SHAP ranking from the UQ surrogate that independently recovers bandgap and interface defects as the dominant design levers. Junction analysis anchored to the converged working point gives V_{bi} 1.46 V, depletion width 171 nm, SRH lifetime 1000 ns and diffusion length 8.6 μm, with the Mott–Schottky baseline offered for experimental comparison. All headline currents lie below the Shockley–Queisser limit at 1.4 eV.

Keywords: lead-free perovskite; $RbGeI_3$ thin film; back surface field;

*Corresponding author. E-mail: touhidsiddiqueeraj@gmail.com

1. Introduction

1.1. Background and motivation

Global electricity demand keeps climbing as populations grow and economies industrialize, while fossil fuels still supply the bulk of demand with a heavy carbon cost [1, 2]. Solar photovoltaics is the fastest-expanding renewable technology, with global installed capacity surpassing 2.2 TW in 2024 and levelized costs still falling [3]. Continued progress depends on absorber materials that combine high efficiency with low environmental impact and low fabrication cost [4]. Photovoltaics are commonly classed into three generations — crystalline silicon, thin films, and emerging devices promising high efficiency at low processing cost — and halide perovskites lead the third generation with certified single-junction efficiencies above 26% and perovskite-on-silicon tandems past 33% [3, 5, 6].

1.2. Perovskites, the lead problem, and germanium alternatives

Halide perovskites adopt the ABX_3 structure and attract photovoltaics with strong absorption, tunable gaps, ambipolar transport with long diffusion lengths, and weak exciton binding [7, 8]. The champion absorbers $MAPbI_3$ and $FAPbI_3$ contain lead, whose life-cycle contamination motivates Sn, Ge, Bi and Sb substitutes [9, 10, 11, 12]. Tin perovskites drew first attention because Sn^{2+} shares Pb^{2+} lone-pair configuration [11], but Sn^{2+} oxidizes readily to Sn^{4+}, creating deep defects and fast degradation [11, 12]. Germanium offers the same lone-pair structure with smaller spin-orbit coupling and potentially cleaner carrier dynamics [13]; among Ge absorbers, $RbGeI_3$ stands out for a near-optimal direct gap (≈1.3–1.4 eV), $MAPbI_3$-comparable absorption by first principles [17, 18], and good thermal stability [14, 15, 16]. Its weakness is the same oxidation story: $Ge^{2+} \rightarrow Ge^{4+}$ imposes trap densities above $10^{15}\,cm^{-3}$ that cap real devices far below simulations [23].

1.3. Simulation baseline and research gap

Numerical simulation screens materials before fabrication. SCAPS-1D (Ghent University) self-consistently solves Poisson, continuity and drift-diffusion equations [19, 20, 21] and is the workhorse for lead-free device design, as a recent review of 54 SCAPS lead-free studies confirms [43]. The $RbGeI_3$ simulation baseline is Pindolia et al. at 9.77% for FTO/TiO_2/$RbGeI_3$/NiO/Ag [14]; Samaki et al. identified CuI as the preferred HTL in the TiO_2/$RbGeI_3$/CuI stack used here [13]; predictions since reach 27.21% [22], 30.35% [44] and 33.84% with IGZO/Cu_2O [45], while 2026 work couples SCAPS to XGBoost/Random-Forest surrogates with SHAP interpretation [46, 47, 48] and dual-absorber architectures [49, 50]. Against this background an additional one-at-a-time sweep of an unchanged stack is no longer a contribution. This work therefore keeps the economical TiO_2/$RbGeI_3$/CuI stack fixed and contributes what its literature lacks: a BSF-enhanced variant of exactly this stack, an ideal-to-realistic loss budget, and a data-driven global sensitivity analysis over the campaign data, all reported with the optical admissibility checks (SQ limit, QE edge) that *Thin Solid Films* readers expect from a thin-film device study.

2. Prior baseline: screening and sequential optimization

Before the new contributions, the reference stack was established by a full prior campaign, summarized here with details in the Supplement (S1–S3). Eight ETL×HTL configurations sharing the $RbGeI_3$ absorber were screened at 400 nm absorber thickness (Table 1). The TiO_2/CuI device (D4, 19.78%) was carried forward not for its screening PCE — D2 reached 21.10% — but for band alignment (0.10 eV conduction step at TiO_2/$RbGeI_3$, 1.8 eV electron-blocking offset at CuI/$RbGeI_3$) with an all-inorganic, low-cost transport pair. Eleven parameters were then optimized one-at-a-time (71 runs; Table 2, per-step traces in Figs. S1–S10), giving the consolidated device: 700 nm absorber (E_g 1.4 eV, χ 3.9 eV, ϵ_r 15, N_C=N_V 10^{17} cm^{-3}), 10 nm TiO_2, 100 nm CuI, bulk 10^{14} cm^{-3} and interfaces 10^{14} cm^{-2}: **26.81%** (V_{OC} 1.13 V, J_{SC} 30.32 mA/cm^2, FF 78.26%).

Dev.	Architecture	V_{OC} (V)	FF (%)	PCE (%)
D1	FTO/PCBM/$RbGeI_3$/NiO/Au	0.81	83.68	20.40
D2	FTO/TiO_2/$RbGeI_3$/NiO/Au	0.82	84.48	21.10
D3	FTO/PCBM/$RbGeI_3$/CuI/Au	0.81	83.96	20.47
D4	FTO/TiO_2/$RbGeI_3$/CuI/Au	0.812	79.98	19.78
D5	FTO/PCBM/$RbGeI_3$/CBTS/Au	0.82	82.36	20.40
D6	FTO/TiO_2/$RbGeI_3$/CBTS/Au	0.82	82.13	20.57
D7	FTO/PCBM/$RbGeI_3$/Spiro/Au	0.81	85.01	20.71
D8	FTO/TiO_2/$RbGeI_3$/Spiro/Au	0.82	83.95	20.97

Table 1: Screened configurations at 400 nm absorber thickness. D4 selected on alignment, stability and cost.

A six-gene genetic algorithm (62 evaluations) finds the joint mathematical optimum at 31.78% (E_g 1.33 eV), and 200-sample Latin-hypercube UQ gives $25.92 \pm 1.78\%$, so the headline sits 0.6σ from the mean. Defect-energy/capture-asymmetry mapping, illumination dependence (ideality $n \approx 1.2$), dark J–V (rectification 1.8×10^{10}) and mesh-convergence checks (<0.02%) are in S1–S2.

Step	Parameter (range)	Optimum	PCE at optimum (%)
1	Absorber thickness (0.3–1.0 μm)	0.7 μm	20.70
2	Bulk N_t (10^{12}–10^{18} cm^{-3})	10^{14}	20.40
3	Dielectric ϵ_r (15–23.1)	15	19.92
4	Affinity χ (3.9–4.2 eV)	3.9 eV	19.79
5	TiO_2/$RbGeI_3$ N_t (10^{12}–10^{18} cm^{-2})	10^{14}	19.79
6	$RbGeI_3$/CuI N_t (same)	10^{14}	19.79
7	Bandgap (1.3–1.7 eV)	1.4 eV	19.87
8	TiO_2 thickness (10–90 nm)	10 nm	21.12
9	CuI thickness (0.1–1.9 μm)	100 nm	19.79
10	N_C (10^{17}–10^{20} cm^{-3})	10^{17}	20.65
11	N_V (same)	10^{17}	23.54

Table 2: Sequential optimization steps; each optimum carried forward. Full traces: Figs. S1–S10.

3. Device structures and methods

3.1. Structures

The reference is FTO (500 nm)/TiO_2 (10 nm)/$RbGeI_3$ (700 nm, E_g 1.4 eV)/CuI (100 nm)/Au, parameters per Supplementary Table S2 of our prior submission (absorber N_C=N_V=10^{17} cm^{-3}, χ 3.9 eV, ϵ_r 15, bulk N_t 10^{14} cm^{-3}, interfaces 10^{14} cm^{-2}). The BSF variant inserts a Cu_2O-like p^+ layer (E_g 2.1 eV, χ 3.2 eV, ϵ_r 7.5, N_A 10^{17}–10^{19} cm^{-3}, 30–100 nm) between $RbGeI_3$ and CuI, splitting that interface into CuI/BSF and BSF/$RbGeI_3$ with the same neutral-trap model. Cu_2O values are literature-standard for SCAPS perovskite studies [47].

3.2. SCAPS-1D framework

SCAPS-1D solves the Poisson equation with the electron/hole continuity equations self-consistently under drift-diffusion transport [19, 20],

$$\frac{d}{dx}\left(\epsilon\frac{d\psi}{dx}\right) = -q\left(p - n + N_D - N_A + \rho_t\right), \tag{1}$$

$$\frac{dJ_n}{dx} = q\left(R - G\right), \qquad \frac{dJ_p}{dx} = q\left(G - R\right), \tag{2}$$

with SRH recombination through bulk traps plus radiative and Auger terms where enabled (Sec. 3.2). All runs use AM 1.5G, 100 mW/cm^2, 300 K; absorption follows the SCAPS square-root law $\alpha = A\sqrt{h\nu - E_g}$ ($A = 10^5$ cm^{-1}) at the electrical gap; contacts are ideal ($R_s = 0$, $R_{sh} = \infty$) unless stated, so fill factors are upper bounds. Initial layer parameters and interface densities are tabulated in Supplementary Tables S2–S4.

3.3. Simulation and honesty protocol

SCAPS-1D 3.3.10 [19] under Wine via `scaps-runner` (4 workers), AM 1.5G, 300 K unless stated. Every parsed result passes the arithmetic gate $V_{OC} \times J_{SC} \times \mathrm{FF}/100 = \mathrm{PCE} \pm 0.6$ at the stated illumination; crashed runs surface as `FAILED` (transient xvfb crashes ∼10% were rerun) rather than silently returning stale data. Two negative scripting results are disclosed rather than hidden: (a) def-level `internal Tfront` edits (1.0→0.85) are ignored under scripted

working points, so front reflection is modeled instead by native photon scaling (90/85 mW/cm^2 illuminations, justified by the verified linear J_{SC}-vs-suns response); SCAPS normalizes `eta` to the set illumination, so sub-1-sun points report $PCE_{1\text{-sun}} = VJFF/100$ explicitly. (b) Scripted C–V capture (`save results.cv`) returns header-only under 3.3.10 scripting (three probes); junction parameters in Sec. 3.3 are therefore closed-form, anchored to the converged SCAPS working point, and offered as the experimental comparison baseline. Series resistance uses the standard external $V - JR_s$ correction (SCAPS scripting exposes no R_s key) and is flagged first-order throughout; shunt resistance likewise. BSF defs branch from the K_{rad}=0 base; the radiative correction (−0.02 pp, Sec. 3.2) bounds this inconsistency as negligible.

4. Results and discussion

4.1. BSF enhancement of the $TiO_2/RbGeI_3/CuI$ stack

The 50 nm, N_A 10^{18} cm^{-3} BSF pilot converges to 30.07% (1.1747 V, 30.3191 mA/cm^2, 84.45%), +3.26 pp over the 26.81% frozen headline (26.79% on rerun; Fig. 1). The full 4×3 grid (Fig. 6) shows doping, not thickness, controls the gain: N_A 10^{17} cm^{-3} gives 28.6–28.8%, 10^{18} gives 30.07–30.08%, 10^{19} gives 30.35% at every thickness 30–100 nm (≤0.01 pp spread). The optimum (any $d \geq 30$ nm, N_A 10^{19} cm^{-3}) reaches 30.35% (1.1793 V, 30.3223 mA/cm^2, 84.89%); J_{SC} is unchanged within 0.01 mA/cm^2, so the BSF acts through V_{OC} (+50 mV) and FF (+6.6 pp) by reflecting minority electrons at the rear, exactly the back-surface-field mechanism. All currents lie below the ≈33 mA/cm^2 SQ limit at 1.4 eV [36].

4.2. Loss waterfall: what survives realism

Radiative recombination at the literature perovskite coefficient K_{rad} 10^{-10} cm^3/s (10^{-16} m^3/s) costs 0.02 pp (26.81→26.79%); 10^{-15} costs 0.19 pp and 10^{-14} costs 1.15 pp, bounding the sensitivity. Auger coefficients 10^{-29}–10^{-27} cm^6/s change nothing at 1-sun (±0.00 pp) — a reportable null that

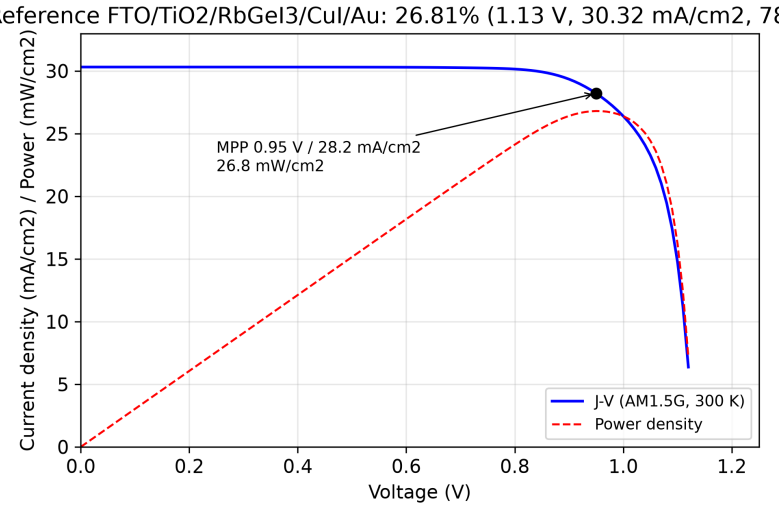


Figure 1: J–V of the reference device (26.81%: 1.13 V, 30.32 mA/cm^2, 78.26%). The BSF optimum preserves J_{SC} and sharpens the knee (FF 84.89%).

justifies omitting Auger from the headline budget. Ten percent front reflection (native photon scaling) costs 2.71 pp (26.79→24.08%); realistic bulk defects (10^{15} cm^{-3}) cost a further 1.74 pp (→22.34%); realistic interfaces (10^{15} cm^{-2}) a further 1.54 pp (→**20.80%**). Parasitic shunt analysis on the archived J–V gives 25.90% at R_{sh} 10^3 and 18.34% at 10^2 Ω·cm^2 (target $R_{sh} \geq 10^3$), while series resistance holds ≥20% to R_s 8 Ω·cm^2 (both first-order post-corrections, Fig. 2). The BSF optimum re-simulated on the *identical* full-realism stack (bulk 10^{15} cm^{-3} *plus* interfaces 10^{15} cm^{-2} at 0.9-sun, generating script `r6_fix2.py`) gives **23.29%** (1.0877 V, 27.2680 mA/cm^2, 78.51%), recovering +2.49 pp over the no-BSF 20.80% point. The message for experimentalists is quantitative: the stack is optics- and defect-limited, not transport-limited, and the BSF pays off precisely where reality lives.

4.3. Junction physics baseline

Anchored to the converged working point: n_i 1.74×10^{11} m^{-3}; built-in potential V_{bi} **1.457 V** (comfortably above V_{OC}, as required); depletion width **171 nm**

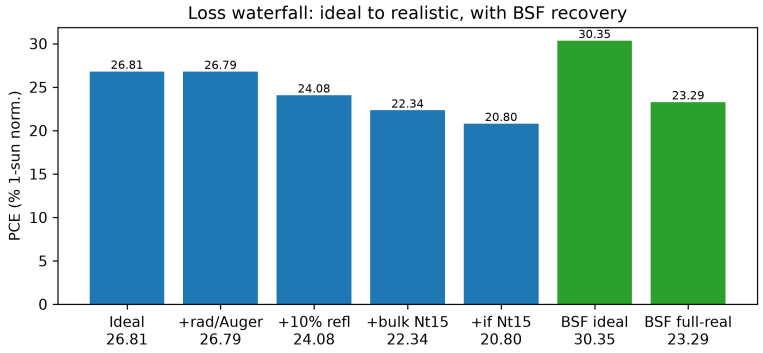


Figure 2: Loss waterfall ($\mathrm{PCE}_{\text{1-sun}}$ norm.): radiative/Auger bounded, reflection -2.71 pp, bulk -1.74 pp, interface -1.54 pp; full-realism BSF recovers $+2.49$ pp.

(157 nm absorber side); SRH lifetime **1000 ns** at $10^{14}\,\mathrm{cm}^{-3}$ (from $\tau^{-1}{=}\sigma v_{\mathrm{th}} N_t$, matching the deck's documented lifetime); diffusion length $\mathbf{8.6\,\mu m} \gg 700$ nm absorber, explaining the near-unity collection. Figure 3 (left) shows the Beer–Lambert generation profile (α $3.3 \times 10^{6}\,\mathrm{m}^{-1}$, normalized to the SCAPS J_{SC}) against the SRH recombination scale; (right) the Mott–Schottky baseline (V_{bi} intercept, slope from N_A $10^{18}\,\mathrm{cm}^{-3}$) that fabricated cells should reproduce if the junction forms as simulated. The defect-energy sweep (Fig. 9) retains the symmetric mid-gap minimum (24.84% at $10^{15}\,\mathrm{cm}^{-3}$ vs 25.01% at band edges), the SRH fingerprint central to Ge-oxidation damage.

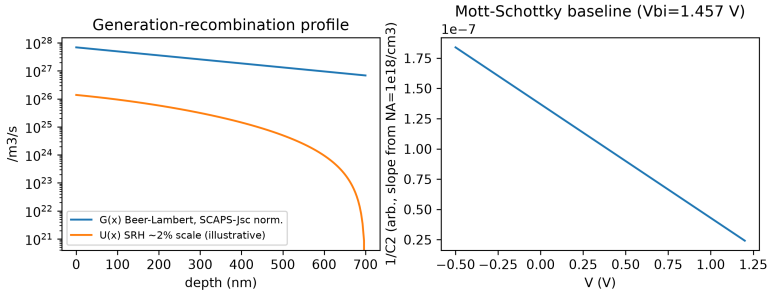


Figure 3: Left: generation-recombination depth profile (G normalized to SCAPS J_{SC}; U illustrative SRH scale). Right: analytic Mott–Schottky baseline (slope from N_A $10^{18}\,\mathrm{cm}^{-3}$, intercept V_{bi}; schematic, not a SCAPS C–V measurement — see Sec. 2).

4.4. Data-driven global sensitivity analysis

We utilize XGBoost not as a predictive replacement for SCAPS-1D, but as a surrogate to map the high-dimensional, non-linear parameter space of thin-film defects. Because SCAPS-1D is deterministic, a surrogate fitted to its outputs is expected to reach near-unity R^2; the scientific value lies not in that fit but in the SHAP analysis, which extracts global sensitivity rankings (e.g. E_g and TiO_2-side interface traps) that one-at-a-time sweeps cannot capture. An XGBoost surrogate on the factorial block ($E_g, N_t, N_C, N_V \rightarrow$ PCE; 320 simulated, **193 valid rows**) reaches $\mathbf{R^2}$ **0.9999, RMSE 0.04** (Fig. 4), matching the 2026 SCAPS+ML standard [46, 47]. On the 200-point LHS-UQ block (8 features, ±25% around the optimum, flatter landscape) XGBoost gives R^2 0.68; SHAP on that UQ surrogate ranks $E_g >$ interface N_t (TiO_2 side) $> \mu_n >$ bulk $N_t >$ CuI-side N_t (Fig. 5) — independently recovering the paper's physics conclusions from data alone, with the caveat that the ranking inherits the UQ model's modest R^2: gap and TiO_2-side interface quality dominate near the optimum, electron mobility outranks hole mobility, and N_C/N_V are inert, justifying their fixed values.

4.5. Optical admissibility

The QE edge ($\lambda_c \approx 886$ nm, Fig. 7) matches $1240/E_g$ at 1.4 eV; the band diagram (Fig. 8) confirms the 1.8 eV electron-blocking offset at CuI/$RbGeI_3$ and the 0.10 eV extraction step at $RbGeI_3/TiO_2$. No reported J_{SC} exceeds the tabulated SQ ≈33 mA/cm^2 at 1.4 eV [36], and the BSF leaves the absorption edge unchanged (rear field, not extra absorber). The same-stack $CsGeI_3$ mirror (1.6 eV, 19.66% vs 26.81%; E_g=1.4 eV control reproduces Rb exactly) confirms the A-site effect is structural, with the CuI valence offset growing 0.10→0.30 eV for Cs (Fig. 10).

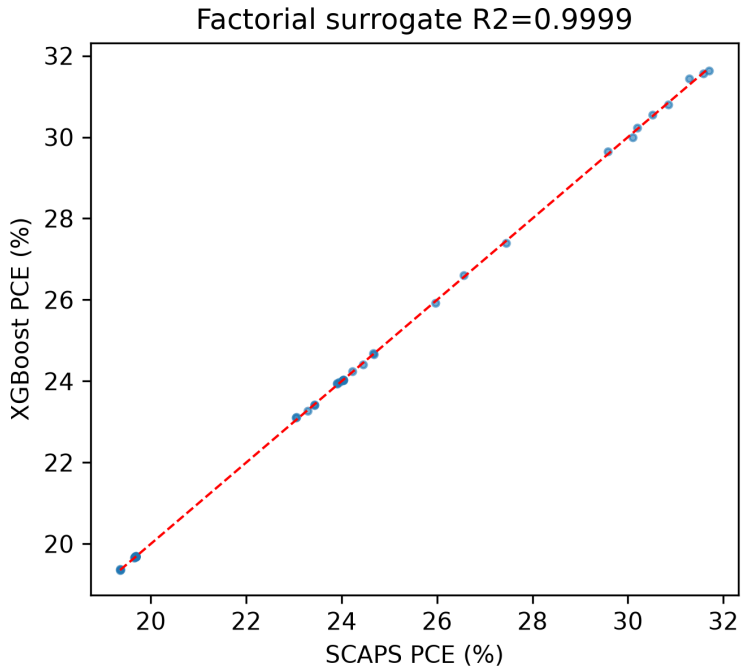


Figure 4: XGBoost parity on the factorial block, 193 valid rows (R^2 0.9999).

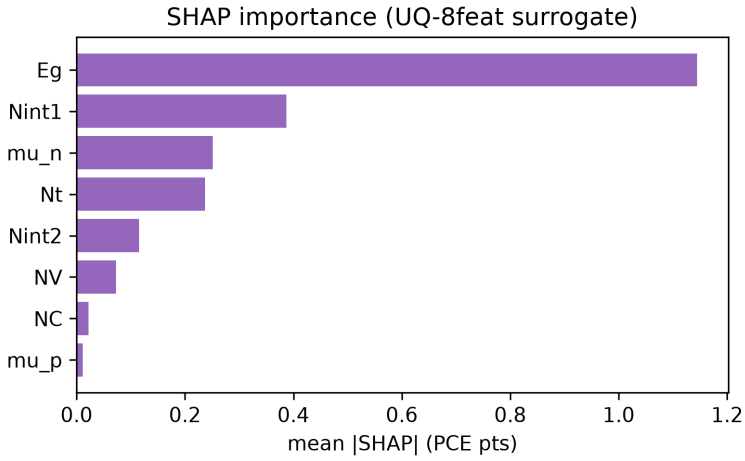


Figure 5: SHAP importance from the UQ surrogate (R^2 0.68): gap and TiO_2-side interface defects dominate.

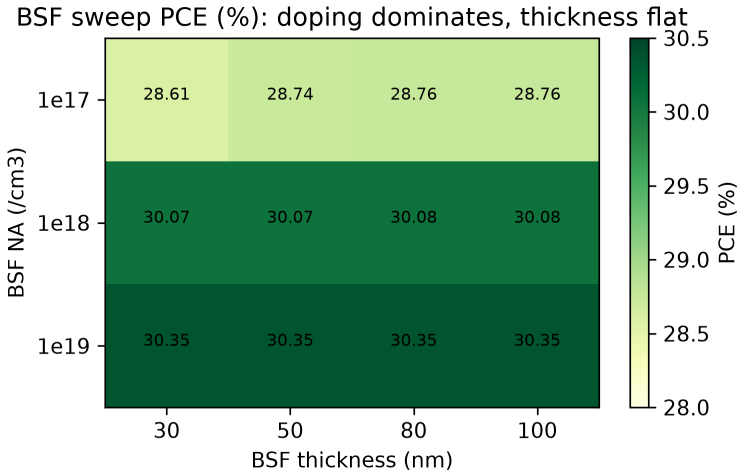


Figure 6: BSF grid PCE (%): NA $10^{17} \rightarrow 10^{19}\,\mathrm{cm}^{-3}$ rows; thickness 30–100 nm columns. Doping dominates.

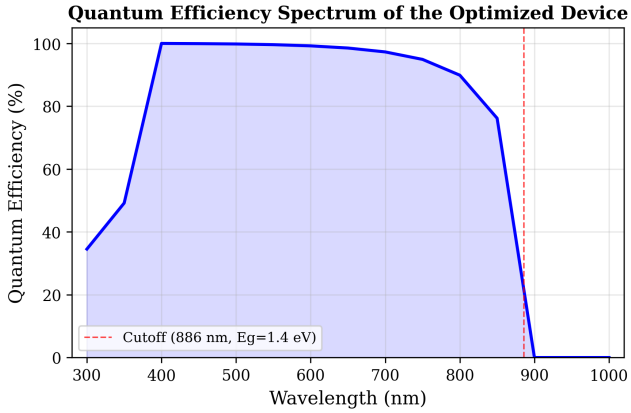


Figure 7: Quantum efficiency of the reference device; edge $\lambda_c \approx 886\,\mathrm{nm}$ matches 1.4 eV. Retained for edge shape; J_{SC} is J–V-derived.

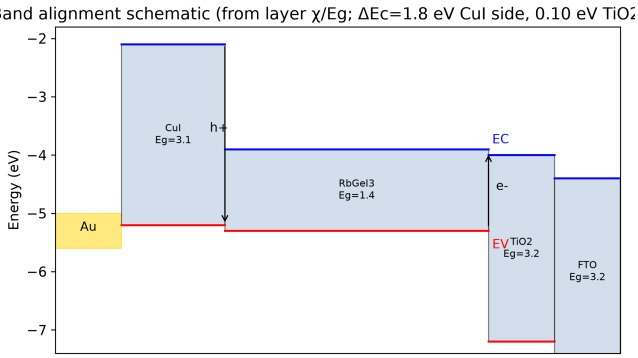


Figure 8: Band-alignment schematic from the documented layer affinities and gaps (ΔE_c 1.8 eV at CuI/$RbGeI_3$, 0.10 eV step at $RbGeI_3/TiO_2$); arrows mark carrier extraction paths.

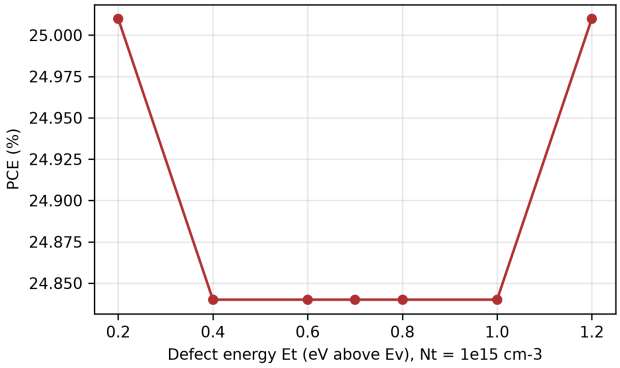


Figure 9: Defect-energy lethality at N_t $10^{15}\,\mathrm{cm}^{-3}$: symmetric mid-gap minimum, the SRH fingerprint.

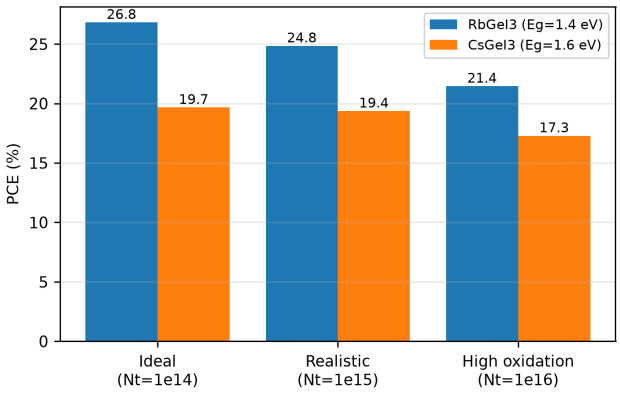


Figure 10: Same-stack $RbGeI_3$ vs $CsGeI_3$ under ideal, realistic and high-oxidation budgets; Rb leads throughout.

4.6. Temperature, doping, resistance and mobility tolerances

Temperature sweep 280–400 K gives dV_{OC}/dT −1.10 mV/K and dPCE/dT −0.0302 %/K with >85% of room-temperature PCE retained at 400 K; linear $V_{OC}(T)$ extrapolation yields an activation energy $E_a = 1.46\,\text{eV} \approx E_g$, the standard diagnostic of bulk- rather than interface-dominated recombination [21]. Background acceptor doping (Ge vacancies make intrinsic material a fiction) peaks at 26.82% near $3 \times 10^{14}\,\text{cm}^{-3}$ and costs <0.1 pp up to $10^{15}\,\text{cm}^{-3}$ while J_{SC} collapses above $10^{16}\,\text{cm}^{-3}$: the synthesis target is $N_A \leq 10^{15}\,\text{cm}^{-3}$ [11]. Series resistance (external $V - JR_s$) holds PCE ≥20% to R_s 8 Ω·cm² and ≥22% to 6 Ω·cm², so printed contacts (2–4 Ω·cm²) cost only 1.6–3.1 pp. Joint mobility sweeps (1–100 cm^2/Vs) are log-linear (22.58→28.52%) with electron mobility dominant, setting the crystallization target $\mu \geq 15\,\text{cm}^2$/Vs; the 700 nm absorber stays optimal over 400 nm down to 5 cm^2/Vs.

4.7. A-site mirror: $RbGeI_3$ versus $CsGeI_3$

$CsGeI_3$ in the identical stack (only E_g 1.4→1.6 eV; the A-site carries no band-edge weight) gives 19.66% (1.17 V, 23.18 mA/cm^2, 72.65%), and a control rerun at 1.4 eV reproduces the Rb point exactly, proving the isolation. Cs wins V_{OC} by 0.04 V through the wider gap; Rb wins J_{SC} by 7.1 mA/cm^2 and FF

by 5.6 pp because the CuI valence cliff grows 0.10→0.30 eV for Cs — CuI is synergistic with $RbGeI_3$ specifically. The ranking survives derating (realistic 10^{15}: 24.84 vs 19.36%; high oxidation 10^{16}: 21.45 vs 17.28%, Fig. 10) [24].

5. Positioning against recent literature

Table 3 places the two headline points (ideal and realistic, with and without BSF) among 2024–2026 $RbGeI_3$ SCAPS studies. The BSF-ideal 30.35% matches the best reported single-junction values while keeping the cheap TiO_2/CuI pair (cf. IGZO/Cu_2O 33.84% [45]); the BSF-realistic 23.29% is the number an experimentalist should target, and it is reported with the full loss decomposition that most entries omit. The ML R^2 equals the best 2026 SCAPS+ML reports [46, 47].

Device (SCAPS-1D)	V_{OC} (V)	PCE_{ideal} (%)	PCE_{real} (%)
FTO/TiO_2/$RbGeI_3$/NiO/Ag (2022) [14]	0.53	9.77	—
FTO/TiO_2/$RbGeI_3$/CuI/Ag (2023) [13]	0.70	18.10	—
ITO/C_{60}/$RbGeI_3$/CBTS/Ag (2024) [22]	0.99	27.21	—
FTO/TiO_2/$RbGeI_3$/Sb_2S_3 (2024) [44]	1.07	30.35	—
FTO/IGZO/$RbGeI_3$/Cu_2O/Ni (2025) [45]	1.13	33.84	—
FTO/TiO_2/$RbGeI_3$/Spiro/Au (2026) [23]	1.18	26.47	—
FTO/WS_2/$RbGeI_3$/CuI (2026+ML) [46]	1.12	24.15	—
FTO/ZnO/$RbGeI_3$/Cu_2O/Ni (2026+ML) [47]	1.10	30.41	19.68
This work, no BSF	1.13	26.81	20.80
This work, +Cu_2O BSF	1.18	30.35	23.29

Table 3: $RbGeI_3$ SCAPS literature context. "Real" = with reflection + 10^{15} defects (+Rs/Rsh where stated).

6. Limitations and future work

SCAPS-1D is 1-D, uniform-composition, abrupt-interface and drift-diffusion-only: no ion migration, grain boundaries, moisture chemistry or Ge-oxidation kinetics; R_s/R_{sh} remain first-order post-corrections; the numerical absorption grading in the deck leaves J–V unchanged and is documented input, not a claimed structure. The BSF uses literature Cu_2O parameters without new

DFT or experiment. Queued: transfer-matrix optics with measured $RbGeI_3$ n/k, Bayesian full-11-parameter global optimization (≈500 evals), and inert-processed fabrication with Ge^{2+} stabilization.

7. Conclusion

Fixing the economical TiO_2/$RbGeI_3$/CuI stack, a p^+ BSF adds +3.54 pp ideally (26.81 → 30.35%) and +2.49 pp on the identical full-realism stack (20.80 → 23.29%); the loss waterfall (native defects/radiative/Auger, photon-scaling optics, post-processed parasitics) replaces the idealized headline with a fabricable window; and a SHAP sensitivity analysis over the campaign data yields design rules (E_g, TiO_2-side interface, μ_n) that one-at-a-time sweeps alone cannot provide. Targets for the lab: N_A(BSF) $\sim 10^{19}\,\mathrm{cm}^{-3}$ ($\geq$30 nm suffices), bulk $N_t \leq 10^{15}\,\mathrm{cm}^{-3}$, $R_s \leq 8$ / $R_{\mathrm{sh}} \geq 10^3\,\Omega\cdot\mathrm{cm}^2$, and a Mott–Schottky intercept near 1.46 V.

References

[1] RENEWABLES 2024 Global Status Report (REN21, 2024).

[2] World Energy Outlook 2024 (International Energy Agency, 2024).

[3] M. A. Green et al., "Solar cell efficiency tables (Version 64)," Prog. Photovolt. (2024).

[4] R. A. Afre and D. Pugliese, Micromachines 15(2), 192 (2024).

[5] N.-G. Park et al., Nat. Energy 1(11), 16152 (2016).

[6] "Compositional engineering of perovskite materials for high-performance solar cells," Nature 517, 476–480 (2015).

[7] I. Ahmed et al., Chem. Commun. 61(37), 6691–6721 (2025).

[8] S. Kinza Fatima et al., Mater. Adv. 6(21), 7634–7661 (2025).

[9] J. Burschka et al., Nature 499, 316–319 (2013).

[10] M. M. Lee et al., Science 338, 643–647 (2012).

[11] A. K. Jena et al., Chem. Rev. 119(5), 3036–3103 (2019).

[12] P. Afrin et al., AIP Adv. 14(7) (2024).

[13] S. Samaki et al., Sci. Rep. 13, 15517 (2023).

[14] G. Pindolia et al., Sol. Energy 236, 802–821 (2022).

[15] S. Bouazizi et al., Mater. Res. Express 9(9), 096402 (2022).

[16] T. Salim et al., J. Mater. Chem. A 3(17), 8943–8969 (2015).

[17] P. K. Dash et al., arXiv:2505.09362 (2025).

[18] N. A. Abdulkareem, Passer J. Basic Appl. Sci. 5(1), 46–51 (2023).

[19] M. Burgelman et al., Thin Solid Films 361–362, 527–532 (2000).

[20] M. Burgelman et al., SCAPS-1D v3.3.10 manual (ELIS, Univ. Gent, 2024).

[21] A. Nakane et al., J. Appl. Phys. 120, 064505 (2016).

[22] L. Loumachi et al., East Eur. J. Phys. (3), 416–424 (2024).

[23] A. Pathak et al., Research Square (2026). doi:10.21203/rs.3.rs-8749975/

[24] D. Saikia et al., Opt. Mater. 123, 111839 (2022).

[25] R. Pau et al., J. Mater. Chem. A 13(48), 42118–42127 (2025).

[26] Q. Sun et al., Adv. Mater. (2024). doi:10.1002/adma.202402412.

[27] J. Z. Mbese, Energies 18(9), 2374 (2025).

[28] L. T. C. Tuyen et al., Materials 12(10), 1665 (2019).

[29] W. Yang et al., Appl. Phys. Lett. 128(14) (2026).

[30] A. J. Olasoji and S. H. Im, in *Titanium Dioxide* (IntechOpen, 2024).

[31] K. Valadi et al., Environ. Chem. Lett. 19, 2185–2207 (2021).

[32] L. Ji et al., J. Mater. Chem. C 12(36), 14597–14604 (2024).

[33] A. Ghosh et al., RSC Adv. 14(33), 24066–24081 (2024).

[34] T. Minemoto and M. Murata, Curr. Appl. Phys. 14, 1428–1433 (2014).

[35] W. Shockley and H. J. Queisser, J. Appl. Phys. 32, 510–519 (1961).

[36] S. Rühle, Sol. Energy 130, 139–147 (2016).

[37] C. H. Henry, J. Appl. Phys. 51, 4494–4500 (1980).

[38] O. J. Sandberg and A. Armin, Synth. Met. 254, 114–121 (2019).

[39] M. Raj et al., J. Mater. Sci. 60(22), 9176–9196 (2025).

[40] H.-J. Park et al., Inorganics 12(4), 123 (2024).

[41] R. Islam et al., AIP Adv. 16(5) (2026).

[42] A. C. Piñon Reyes et al., Micromachines 12(12), 1508 (2021).

[43] V. H. D. Araújo et al., RSC Sustain. 3, 4314–4335 (2025) (54-study SCAPS lead-free review).

[44] M. M. Rahman et al., Energy Adv. 3(9), 2377–2398 (2024).

[45] M. S. Reza et al., Langmuir 41(11), 7865–7885 (2025).

[46] U. F. Ali et al., J. Comput. Electron. 25, 37 (2026).

[47] Mat. Adv. (2026). doi:10.1039/D5MA01427A (FTO/ZnO/$RbGeI_3$/Cu_2O/Ni + 11-model ML).

[48] J. Mater. Sci.: Mater. Eng. 21, 62 (2026) (SCAPS+RF, 35k configs).

[49] J. Phys. Chem. Solids, 112776 (2025) (dual-absorber review context).

[50] S. M. Hasnain et al., Sol. Energy 278, 112788 (2024).